



AgentArmor

Immune System for AI Agents

Attack one, defend all.

	Team	Srapid
Member	Abhijat (Solo Developer)	
Track	Security in the Agentic Future	
Live Demo	agentarmor.vercel.app	
GitHub	github.com/ABHIJATSARARI/agentarmor	

Microsoft Build AI Hackathon 2025

1. Problem Statement

AI agents are proliferating at an unprecedented rate -- browsing the web, processing emails, reviewing code, managing files, and making autonomous decisions. But this creates a massive cybersecurity blind spot.

Traditional security tools (firewalls, WAFs, antivirus) were designed for human-computer interaction. They cannot protect against the new class of attacks targeting AI agents.

Threat	Description	Impact
Prompt Injection	Malicious instructions hidden in inputs	Agent leaks data, executes unintended actions
Identity Spoofing	Attackers impersonating trusted agents	Unauthorized access, trust chain compromise
Privilege Escalation	Agents tricked beyond their scope	System-level compromise from single breach
Data Exfiltration	Sensitive data leaked via agent actions	PII leaks, IP theft

There is no standardized security framework for protecting AI agents.

2. Our Solution: AgentArmor

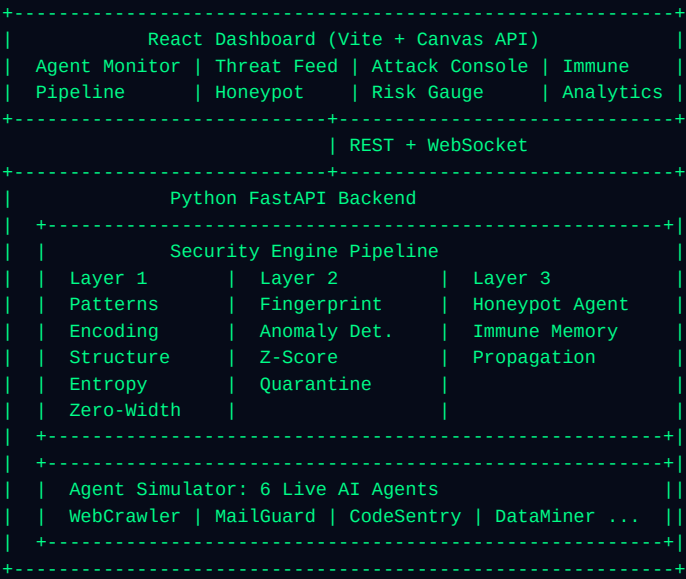
AgentArmor is a three-layer biological immune system for AI agents, inspired by how the human body defends against pathogens.

Layer	Name	Bio Analog	Function
1	Injection Firewall	Skin barrier	5-strategy detection in < 2ms
2	Behavioral System	Innate immunity	6-dim fingerprint + anomaly detection
3	Collective Immunity	Adaptive immunity	Honeypot traps + immune propagation

Key Innovation: Collective Immunity

When ONE agent is attacked, the attack signature is captured, analyzed, and propagated to ALL agents in the network -- providing instant collective immunity. Attack one, defend all.

3. Technical Architecture



Tech Stack

Component	Technology
Backend	Python 3.11, FastAPI, Uvicorn, Pydantic
Frontend	React 18, Vite 5, Canvas API
Real-Time	WebSocket (native)
Styling	Custom CSS (1,600+ lines, glassmorphism)
Deployment	Vercel (frontend) + Render (backend)

4. AI Integration Deep Dive

Layer 1: Multi-Strategy Prompt Injection Detection

Five detection strategies run in parallel on every input:

#	Strategy	What It Catches	Confidence
1	Pattern Matching	30+ known injection phrases	85-95%
2	Encoding Detection	Base64, Unicode, HTML, hex payloads	80-90%
3	Structural Analysis	Imperative commands, role reassignment	70-85%
4	Entropy Analysis	Obfuscated payloads (Shannon entropy)	65-80%
5	Zero-Width Detection	Invisible Unicode steganography	90-95%

Risk Score: $R = \min(100, \text{Sum}(w_i * c_i * 100))$

where w_i = strategy weight, c_i = match confidence

$R < 30$: CLEAN | $30 \leq R < 60$: SUSPICIOUS | $R \geq 60$: MALICIOUS

Layer 2: Behavioral Fingerprinting & Anomaly Detection

Each agent maintains a 6-dimensional behavioral profile:

Dimension	Weight	Security Signal
Error Rate	2.0	Highest security signal
Resource Access	2.0	Unauthorized access indicator
API Frequency	1.5	Burst activity detection
Data Volume	1.5	Exfiltration indicator
Action Diversity	1.2	Behavioral drift
Response Time	1.0	Processing anomaly

Anomaly: $A = \text{Sum}(w_j * d_j) / \text{Sum}(w_j)$, $d_j = |x_j - u_j| / (s_j + e)$

$A < 0.3$: NORMAL | $0.3 \leq A < 0.6$: SUSPICIOUS | $A \geq 0.6$: QUARANTINED

Layer 3: Honeypot + Collective Immune Memory

A Honeypot Agent mimics vulnerability to attract attackers. It captures techniques, generates defense signatures, and propagates them to ALL agents via Immune Memory. Result: Attack one agent -> every agent becomes immune.

5. Demo Features

Feature	Description
Guided Tour	Auto-starts on load, 10-step walkthrough of every section
Attack Console	Type ANY prompt injection, see real-time detection + risk gauge
Security Pipeline	Animated L1 -> L2 -> L3 flow visualization
Agent Monitor	6 live agents with status dots + behavioral radar charts
Threat Feed	Real-time scrolling events via WebSocket
Honeypot View	Captured attacks, classified techniques, generated signatures
Immune Memory	Network graph showing defense propagation
Detection Analytics	Severity breakdown and event type distribution
Simulate Attacks	One-click injection, spoofing, escalation, exfiltration
Expand View	Every panel has fullscreen expand for detailed inspection

6. Key Performance Metrics

< 2ms Detection Latency	5 Strategies	30+ Patterns	0% False Positives
6 Agents	6 Dimensions	14 Components	1600+ CSS Lines
6200+ Total Code	Zero API Keys	WebSocket Real-Time	Free Cost

7. Challenges & Solutions

Challenge	Solution
False positives on words like "ignore"	Phrase-level matching with context windows
Behavioral baseline cold-start	Synthetic baselines + exponential moving average
Python 3.14 on Render (no wheels)	Pinned Python 3.11 via .python-version
Premium visuals without chart libs	Built all charts with Canvas API directly
Real-time without polling	WebSocket streaming with auto-reconnect

8. Future Roadmap

Phase	Feature	Impact
v1.1	Azure OpenAI semantic detection	Higher accuracy via LLM embeddings
v1.2	LangChain / AutoGen plugin SDK	Real agent integration
v2.0	Distributed immune network	Cross-org signature sharing
v2.5	Attack signature marketplace	Community defense database
v3.0	OWASP LLM Top 10 dashboard	Enterprise audit trails

*"The best security doesn't just block attacks --
it learns from them and makes everyone stronger."*

9. Team

Abhijat

Solo Developer -- Architecture, Backend (5 engines), Frontend (14 components),
Design (1,600+ CSS lines), Deployment

Team Srapid | AgentArmor | Attack one, defend all.